

v1.0 | Validation Report

FlowForge

1D Meanline Centrifugal Pump Preliminary Design Solver

Tool Version	FlowForge_v1_0.py / FlowForge_v1_0.html
Standard References	Güllich (2020) · Dixon & Hall (2014) · Pfeiderer (1961) · Wiesner (1967) · NIST
Scope	3 Scenarios — Wiesner / Pfeiderer / Stodola · Design Point + Performance Curve

This document covers the engineering validation of the FlowForge v1.0 centrifugal pump preliminary design solver. Three independent scenarios are executed; analytical reference calculations are carried out step by step and compared against solver outputs. In all scenarios the solver results are in full agreement with the independently verified theoretical values. The tool correctly computes impeller geometry, velocity triangles, hydraulic losses, disk friction, slip factors, and specific speed across three distinct slip models.

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SCENARIO 1

Wiesner Slip Model — Industrial Water Pump (CHECK Mode)

This scenario validates the core design solver using the Wiesner (1967) slip model. Input values represent a standard medium-capacity industrial process pump operating at 50 Hz (2-pole motor). A fixed-km architecture is used; results are compared step by step against independent manual calculations.

1.1 Input Parameters

Parameter	Value
Flow rate Q	0.05 m³/s
Target head H	40 m
Rotational speed n	2900 rpm
Outlet blade angle β_2	25°
Flow coefficient km	0.15
Blade count Z	7
Slip model	Wiesner (1967)
Fluid density ρ	998 kg/m³ (water @ 20°C)
Overall efficiency η	0.75 (user override)
Inlet blade angle β_1 (design)	20°
Tip speed limit u_{2_max}	200 m/s
Blockage factor ε	0.08

1.2 Analytical Calculation Steps

Step	Expression	Result
1	$u_{est} = \sqrt{g \cdot H} = \sqrt{9.81 \times 40}$	19.81 m/s
2	$D_{2_init} = 60 \cdot u_{est} / (\pi \cdot n) = 60 \times 19.81 / (\pi \times 2900)$	130.4 mm
3	$\phi_{lim} = (0.7/Z)^{[1/(1+\sin\beta_2)]} = (0.7/7)^{[1/(1+\sin25^\circ)]}$	0.1982
4	$km = 0.15 < \phi_{lim} = 0.1982 \rightarrow$ Wiesner valid	check OK
5	$\sigma_{Wiesner} = 1 - \sqrt{\sin\beta_2} / Z^{0.7} = 1 - \sqrt{\sin25^\circ} / 7^{0.7}$	0.8335
6	u_2 [converged D_2 , iteration 7]	26.545 m/s
7	$V_{m_2} = km \cdot u_2 = 0.15 \times 26.545$	3.982 m/s
8	$V_{u_2_th} = u_2 - V_{m_2}/\tan\beta_2 = 26.545 - 3.982/\tan25^\circ$	17.415 m/s
9	$H_{euler} = \sigma \cdot u_2 \cdot V_{u_2_th} / g = 0.8335 \times 26.545 \times 17.415 / 9.81$	40.612 m
10	$r_2 = D_2/2 = 87.41$ mm; $b_2 = Q/[2\pi \cdot r_2 \cdot V_{m_2} \cdot (1-\varepsilon)]$	24.80 mm
11	$H_{hyd} = k_h \cdot V_{m_2}^2 / (2g) \cdot (D_2/b_2),$ $k_h = 0.15$	0.557 m
12	$D_1 = 0.5 \cdot D_2 = 87.41$ mm; $b_1 = Q/(\pi \cdot D_1 \cdot V_{m_1_target})$	13.00 mm

13	$\beta_{1_flow} = \text{atan}(V_{m1}/u_1) = \text{atan}(3.500/13.34)$	14.77°
14	$i_{eff} = \beta_{1_design} - \beta_{1_flow} = 20^\circ - 14.77^\circ $	5.23°
15	$H_{shock} = k_i \cdot i_{eff_rad}^2 \cdot u_1^2 / (2g), \quad k_i = 0.5$	0.333 m
16	$H_{net} = H_{euler} - H_{hyd} - H_{shock} = 40.612 - 0.557 - 0.333$	39.722 m
17	$\text{convergence error} = 40.000 - 39.722 / 40.000 \times 100$	0.695 %
18	$\omega = 2\pi \cdot n / 60 = 2\pi \times 2900 / 60$	303.7 rad/s
19	$P_{df} = k_{df} \cdot \rho \cdot \omega^3 \cdot D_2^5 \cdot (1 + 5 \cdot b_2 / D_2), \quad k_{df} = 1.5 \times 10^{-3}$	11 713 W
20	$P_{useful} = \rho \cdot g \cdot Q \cdot H_{net} = 998 \times 9.81 \times 0.05 \times 39.722$	19 461 W
21	$\eta_h = P_{useful} / (P_{useful} + P_{df}) = 19\,461 / 31\,174$	0.624
22	$\Omega_s = \omega \cdot \sqrt{Q} / (g \cdot H_{net})^{0.75}$	0.7743
23	$n_{s_delivered} = n \cdot \sqrt{Q} / H_{net}^{0.75} = 2900 \cdot \sqrt{0.05} / 39.722^{0.75}$	40.98
24	$n_{s_design} = n \cdot \sqrt{Q} / H_{euler}^{0.75} = 2900 \cdot \sqrt{0.05} / 40.612^{0.75}$	39.90
25	$\Omega_s \times 52.93 \text{ (cross-check)}$	40.98 ✓

1.3 Verification Results

Check	Calculated	Reference / Limit	Status
[1] H_{net} convergence error	0.695 %	Target: 40.000 m	✓ PASS
[2] σ — Wiesner formula	0.8335	Manual: 0.8335 → $\Delta = 0.0000$	✓ PASS
[3] H_{euler}	40.612 m	Manual: 40.612 m → $\Delta < 0.001$	✓ PASS
[4] u_2 (tip speed)	26.545 m/s	Manual: 26.545 m/s → $\Delta = 0.000$	✓ PASS
[5] D_2	174.82 mm	Geometry self-consistent	✓ PASS
[6] $b_2/D_2 \in [0.02, 0.25]$	0.1422	Hard-validation range	✓ PASS
[7] P_{df}	11 713 W	Manual: 11 713 W → $\Delta = 0.000$	✓ PASS
[8] Ω_s	0.7743	Manual: 0.7743 → $\Delta < 0.001$ %	✓ PASS
[9] $\Omega_s \times 52.93 \approx n_{\text{s_delivered}}$	40.98	Manual: 40.98 → $\Delta < 0.01$ %	✓ PASS
[10] Iterations to convergence	7	Within max_iter = 200	✓ PASS
[11] $n_{\text{s_design}} \in [10, 80]$	39.90	Hard-validation range	✓ PASS
[12] slip_model_used = wiesner	wiesner	No fallback triggered	✓ PASS

SCENARIO 1 RESULT ✓ FULL AGREEMENT WITH ANALYTICAL REFERENCE

SCENARIO 2

Pfleiderer Slip Model — HVAC Cooling Pump, 50% EG/Water (CHECK Mode)

This scenario validates the Pfleiderer (1961) slip model and the EG/water density table lookup. A low-speed pump for a commercial building HVAC circuit is analysed. Flow rate is supplied in m³/h to exercise the unit-conversion path. Mixture density is retrieved from the internal lookup table at 50% EG concentration and 20°C.

2.1 Input Parameters

Parameter	Value
Flow rate Q	180 m ³ /h → 0.050000 m ³ /s
Target head H	22 m
Rotational speed n	1450 rpm
Outlet blade angle β_2	30°
Flow coefficient k_m	0.18
Blade count Z	6
Slip model	Pfleiderer (1961)
Fluid	50% EG / Water → $\rho = 1082$ kg/m ³ (table lookup @ 20°C)
Overall efficiency η	0.75 (user override)
Inlet blade angle β_1 (design)	20°

2.2 Analytical Calculation Steps

Step	Expression	Result
1	$Q_{SI} = 180 / 3600$	0.050000 m ³ /s
2	$\rho \leftarrow EG_RHO_TABLE[50][20]$	1082.0 kg/m ³
3	$\sigma_{Pfleiderer}: p = 0.65 \cdot (1 + \sin\beta_2) = 0.65 \cdot (1 + \sin 30^\circ)$	0.9750
4	$\sigma = 1 / (1 + p \cdot k_m / \tan\beta_2) = 1 / (1 + 0.975 \times 0.18 / \tan 30^\circ)$	0.7669
5	$u_{est} = \sqrt{g \cdot H} = \sqrt{(9.81 \times 22)}$	14.69 m/s
6	$D_{2_init} = 60 \cdot u_{est} / (\pi \cdot n) = 60 \times 14.69 / (\pi \times 1450)$	193.8 mm
7	u_2 [converged, iteration 8]	20.97 m/s → $D_2 = 276.24$ mm
8	$V_{m2} = k_m \cdot u_2 = 0.18 \times 20.97$	3.775 m/s
9	$V_{u2_th} = u_2 - V_{m2} / \tan\beta_2 = 20.97 - 3.775 / \tan 30^\circ$	14.43 m/s
10	$H_{euler} = \sigma \cdot u_2 \cdot V_{u2_th} / g = 0.7669 \times 20.97 \times 14.43 / 9.81$	23.67 m
11	$b_2 = Q / [2\pi \cdot r_2 \cdot V_{m2} \cdot (1 - \varepsilon)]$	36.10 mm
12	$H_{hyd} = k_h \cdot V_{m2}^2 / (2g) \cdot (D_2 / b_2)$	0.505 m
13	H_{shock} (incidence at converged D_1)	1.316 m
14	$H_{net} = H_{euler} - H_{hyd} - H_{shock} = 23.67 - 0.505 - 1.316$	21.848 m

15	convergence error = $ 22.000 - 21.848 / 22.000 \times 100$	0.692 %
16	Q unit conversion check: $180 \text{ m}^3/\text{h} \times (1/3600)$	0.050000 m^3/s ✓

2.3 Verification Results

Check	Calculated	Reference / Limit	Status
[1] Unit conversion ($\text{m}^3/\text{h} \rightarrow \text{m}^3/\text{s}$)	0.050000 m^3/s	Exact: 180/3600	✓ PASS
[2] EG/Water density lookup	1082.0 kg/m^3	Table EG_RHO_TABL E[50][20]	✓ PASS
[3] p — Pfeiderer	0.9750	Manual: $0.65 \times (1 + \sin 30^\circ) = 0.9750$	✓ PASS
[4] σ — Pfeiderer formula	0.7669	Manual: 0.7669 $\rightarrow \Delta = 0.0000$	✓ PASS
[5] H_{net} convergence error	0.692 %	Target: 22.000 m	✓ PASS
[6] D_2	276.24 mm	Geometry self-consistent	✓ PASS
[7] $b_2/D_2 \in [0.02, 0.25]$	0.1307	Hard-validation range	✓ PASS
[8] $H(Q)$ monotone decreasing	Verified	Physical requirement	✓ PASS

SCENARIO 2 RESULT ✓ FULL AGREEMENT WITH ANALYTICAL REFERENCE

SCENARIO 3

Stodola Slip Model — Fire-Fighting Pump, $n_s < 10$ Boundary Test

This scenario intentionally places the design in the $n_s < 10$ regime where a single-stage radial impeller is not the appropriate machine type. The solver mathematics are verified to converge correctly; the hard-validation layer is verified to correctly raise FAIL flags for n_{s_design} and b_2/D_2 . The Stodola formula is also verified as an analytical standalone check.

3.1 Input Parameters

Parameter	Value
Flow rate Q	10 L/s → 0.01000 m³/s
Target head H	80 m
Rotational speed n	3000 rpm
Outlet blade angle β_2	25°
Flow coefficient k_m	0.15
Blade count Z	7
Slip model	Stodola-corrected
Fluid density ρ	983.199 kg/m³ (water @ 60°C – NIST)
Overall efficiency η	0.75 (user override)
Inlet blade angle β_1 (design)	20°
Purpose	Hard-validation boundary test – $n_s < 10$, $b_2/D_2 < 0.02$

3.2 Analytical Calculation Steps

Step	Expression	Result
1	$Q_{SI} = 10 / 1000$	0.01000 m³/s
2	$\sigma_{Stodola} = 1 - (\pi/Z) \cdot \sin\beta_2 = 1 - (\pi/7) \cdot \sin 25^\circ$	0.8103
3	$u_{est} = \sqrt{g \cdot H} = \sqrt{9.81 \times 80}$	28.01 m/s
4	$D_{2_init} = 60 \times 28.01 / (\pi \times 3000)$	178.3 mm
5	u_2 [converged] → D_2	240.20 mm
6	$V_{m2} = k_m \cdot u_2 = 0.15 \times 37.74$	5.661 m/s
7	$V_{u2_th} = u_2 - V_{m2}/\tan\beta_2$	25.61 m/s
8	$H_{euler} = \sigma \cdot u_2 \cdot V_{u2_th} / g = 0.8103 \times 37.74 \times 25.61 / 9.81$	79.97 m
9	$b_2 = Q / [2\pi \cdot r_2 \cdot V_{m2} \cdot (1-\varepsilon)] \rightarrow b_2/D_2$	0.0107 < 0.02
10	H_{net} [converged]	79.71 m
11	convergence error = $ 80.000 - 79.71 / 80.000 \times 100$	0.36 %
12	$n_{s_design} = n \cdot \sqrt{Q} / H_{euler}^{0.75} = 3000 \cdot \sqrt{0.01} / 79.97^{0.75}$	9.92
13	$n_{s_design} = 9.92 < 10 \rightarrow$ outside [10, 80] hard-validation	FAIL – by design

	limit	
14	$b_2/D_2 = 0.0107 < 0.02 \rightarrow$ outside $[0.02, 0.25]$ hard- validation limit	FAIL – by design
15	$\sigma_{\text{Stodola analytical}}$ (standalone): $1 - \pi/7 \cdot \sin 25^\circ$	0.8103 ✓

3.3 Verification Results

Check	Calculated	Reference / Limit	Status
[1] Unit conversion (L/s $\rightarrow$ m³/s)	0.01000 m³/s	Exact: 10/1000	✓ PASS
[2] σ_{Stodola} formula	0.8103	$1 - \pi/7 \cdot \sin 25^\circ = 0.8103$	✓ PASS
[3] H_{net} convergence error	0.36 %	Target: 80.000 m	✓ PASS
[4] $n_{\text{s_design}} < 10$ $\rightarrow$ FAIL flag	9.92	Hard-validation correctly FAIL	✓ PASS
[5] $b_2/D_2 < 0.02 \rightarrow$ FAIL flag	0.0107	Hard-validation correctly FAIL	✓ PASS
[6] Solver math correct despite flags	Converged	Design-space check $\neq$ solver error	✓ PASS

Note: The FAIL flags on $n_{\text{s_design}}$ and b_2/D_2 are the intended and expected outcome for this scenario. The solver mathematics converge correctly; the hard-validation layer correctly identifies that a single-stage radial impeller is outside its applicable design space at this Q/H combination. This confirms that the solver detects critical out-of-range conditions rather than silently accepting an infeasible geometry.

SCENARIO 3 RESULT ✓ SOLVER MATH CORRECT · HARD VALIDATION CORRECTLY
FLAGS OUT-OF-RANGE

SUMMARY AND OVERALL ASSESSMENT

#	Scenario	Mode / Feature	Solver Result	Agreement
1	Industrial Water Pump	Wiesner, CHECK	H _{net} err 0.695 %	✓ FULL
2	HVAC 50% EG/Water	Pfeiderer, CHECK	H _{net} err 0.692 %	✓ FULL
3	Fire-Fighting Pump (ns<10)	Stodola, BOUNDARY	Hard-val. FAIL flags correct	✓ FULL

In all three independent validation scenarios, the FlowForge v1.0 analysis engine demonstrated full agreement with step-by-step analytical reference calculations conforming to Gülich (2020), Wiesner (1967), Pfeiderer (1961) and Dixon & Hall (2014). The solver correctly computes impeller outlet diameter, channel width, velocity triangles, hydraulic and shock losses, disk friction, slip factors for all three implemented models, specific speed (both metric and dimensionless), and hydraulic efficiency. The EG/water density table and flow-unit conversion routines operate without error.

Scenario 3 intentionally triggers the hard-validation FAIL flags, confirming that the solver correctly detects and reports when a design falls outside the applicable range of a single-stage radial impeller. This is an important indicator that the tool does not silently accept infeasible geometries.

OVERALL RESULT ✓ ALL SCENARIOS PASSED — FULL AGREEMENT WITH ANALYTICAL REFERENCE

References

[1]	Gülich, J.F. (2020). Centrifugal Pumps, 3rd Ed. Springer. \$3.3, \$3.6.
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[4]	Dixon, S.L. & Hall, C.A. (2014). Fluid Mechanics and Thermodynamics of Turbomachinery, 7th Ed. Butterworth-Heinemann.
[5]	NIST Webbook — Water thermophysical properties at 0.1 MPa, liquid phase. https://webbook.nist.gov